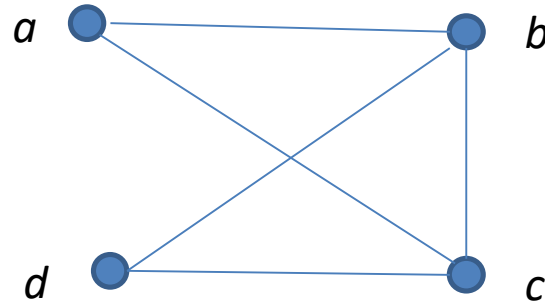


Introduction to Graphs

Graphs

Definition: A *graph* $G = (V, E)$ consists of a nonempty set V of *vertices* (or *nodes*) and a set E of *edges*. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its endpoints.

Example: This is a graph with four vertices and five edges.



Remarks:

We have a lot of freedom when we draw a picture of a graph. All that matters is the connections made by the edges, not the particular geometry depicted. For example, the lengths of edges, whether edges cross, how vertices are depicted, and so on, do not matter

A graph with an infinite vertex set or infinite edges is called an *infinite graph*. A graph with a finite vertex set and finite edges is called a *finite graph*. We restrict our attention to finite graphs.

Some Terminology

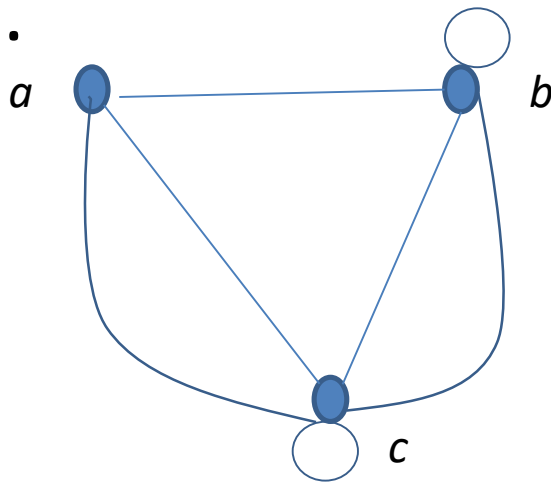
- In a *simple graph* each edge connects two different vertices and no two edges connect the same pair of vertices. (No loops and no parallel edges)
- *Multigraphs* may have multiple edges connecting the same two vertices. When m different edges connect the vertices u and v , we say that $\{u, v\}$ is an edge of *multiplicity* m .

Some Terminology

- An edge that connects a vertex to itself is called a *loop*.
- A *pseudo graph* may include loops, as well as multiple edges connecting the same pair of vertices.
- In a *simple graph* each edge connects two different vertices and no two edges connect the same pair of vertices (No loops and no parallel edges)

Example:

This pseudo graph has both multiple edges and loops.



Directed Graphs

Definition: An *directed graph* (or *digraph*) $G = (V, E)$ consists of a nonempty set V of *vertices* (or *nodes*) and a set E of *directed edges* (or *arcs*). Each edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to *start at* u and *end at* v .

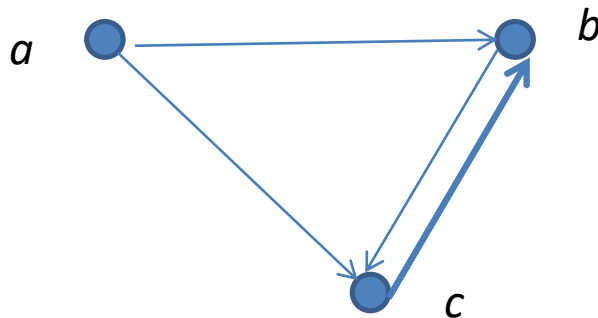
Remark:

- Graphs where the end points of an edge are not ordered are said to be *undirected graphs*.

Let (u,v) be an edge in G . Then u is the *initial vertex* of this edge and is *adjacent to* v and v is the *terminal* (or *end*) *vertex* of this edge and is *adjacent from* u . The initial and terminal vertices of a loop are the same.

Some Terminology (*continued*)

A simple directed graph has no loops and no multiple edges.

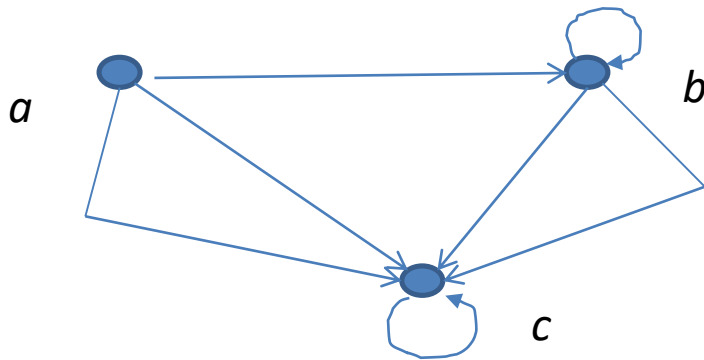


Example:

This is a directed graph with three vertices and four edges.

A *directed multigraph* may have multiple directed edges. When there are m directed edges from the vertex u to the vertex v , we say that (u,v) is an edge of *multiplicity* m .

Example:



In this directed multigraph the multiplicity of (a,b) is 1 and the multiplicity of (b,c) is 2.

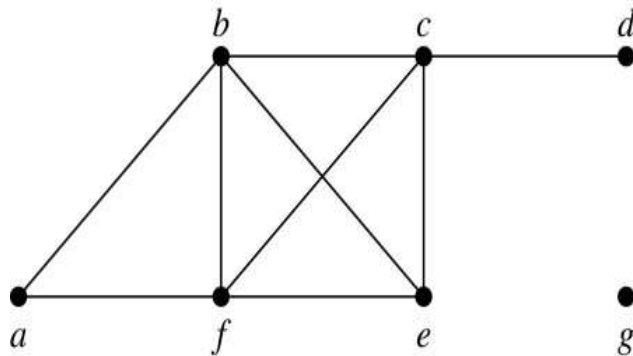
Some Terminology (*continued*)

Definition 1. Two vertices u, v in an undirected graph G are called *adjacent* (or *neighbors*) in G if there is an edge e between u and v . Such an edge e is called *incident with* the vertices u and v and e is said to *connect* u and v .

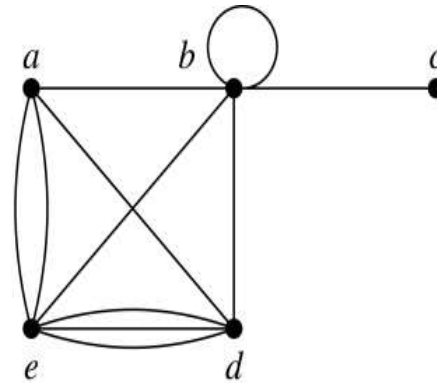
Definition 2. The *degree of a vertex* in a undirected graph is the number of edges incident with it, except that a loop at a vertex contributes two to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Degrees and Neighborhoods of Vertices

Example: What are the degrees and neighborhoods of the vertices in the graphs G and H ?



G



H

Solution:

G : $\deg(a) = 2, \deg(b) = \deg(c) = \deg(f) = 4, \deg(d) = 1,$
 $\deg(e) = 3, \deg(g) = 0.$

$N(a) = \{b, f\}, N(b) = \{a, c, e, f\}, N(c) = \{b, d, e, f\},$
 $N(d) = \{c\}, N(e) = \{b, c, f\}, N(f) = \{a, b, c, e\}, N(g) = \emptyset$

.

H : $\deg(a) = 4, \deg(b) = \deg(e) = 6, \deg(c) = 1, \deg(d) = 5.$

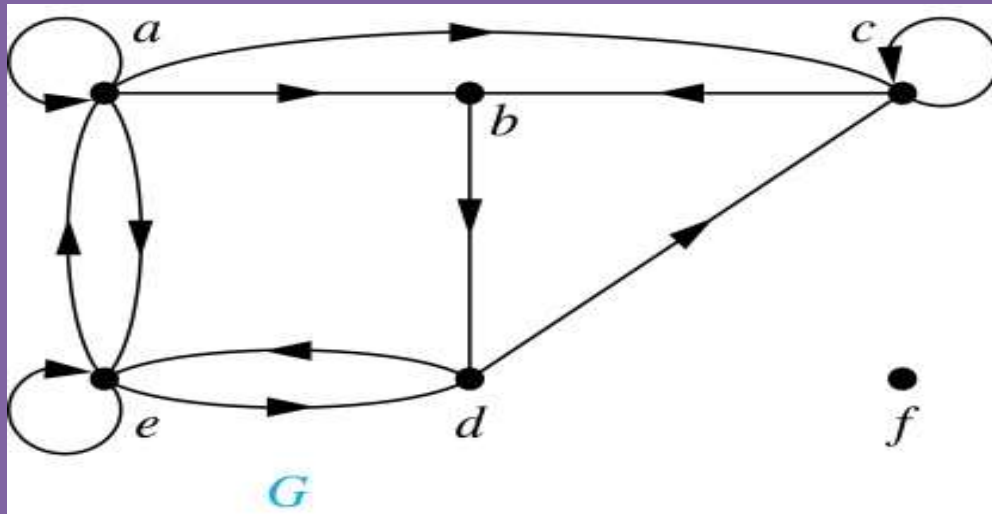
$N(a) = \{b, d, e\}, N(b) = \{a, b, c, d, e\}, N(c) = \{b\},$
 $N(d) = \{a, b, e\}, N(e) = \{a, b, d\}.$

Directed Graphs (*continued*)

Definition: The *in-degree of a vertex v* , denoted $\text{deg}^-(v)$, is the number of edges which terminate at v . The *out-degree of v* , denoted $\text{deg}^+(v)$, is the number of edges with v as their initial vertex. Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of the vertex.

Example:

In the graph G we have



$$\deg^-(a) = 2, \deg^-(b) = 2, \deg^-(c) = 3, \deg^-(d) = 2, \deg^-(e) = 3, \deg^-(f) = 0.$$

$$\deg^+(a) = 4, \deg^+(b) = 1, \deg^+(c) = 2, \deg^+(d) = 2, \deg^+(e) = 3, \deg^+(f) = 0.$$

Directed Graphs (*continued*)

Theorem 3: Let $G = (V, E)$ be a graph with directed edges. Then:

$$|E| = \sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v).$$

Proof: The first sum counts the number of outgoing edges over all vertices and the second sum counts the number of incoming edges over all vertices. It follows that both sums equal the number of edges in the graph.

Theorem 1 (*Handshaking Theorem*): If $G = (V, E)$ is an undirected graph with e edges, then $2e = \sum_{v \in V} \deg(v)$

Proof:

Each edge contributes twice to the degree count of all vertices. Hence, both the left-hand and right-hand sides of this equation equal twice the number of edges.

Handshaking Theorem

Example: How many edges are there in a graph with 10 vertices of degree six?

Solution: Because the sum of the degrees of the vertices is $6 \cdot 10 = 60$, the handshaking theorem tells us that $2m = 60$. So the number of edges $m = 30$.

Example: If a graph has 5 vertices, can each vertex have degree 3?

Solution: This is not possible by the handshaking theorem, because the sum of the degrees of the vertices $3 \cdot 5 = 15$ is odd.

Degree of Vertices (*continued*)

Theorem 2: *The number of vertices with odd degree is always even.*

Proof: Let V_1 be the vertices of even degree and V_2 be the vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. Then

$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

must be even
since $\deg(v)$
is even for
each $v \in V_1$

This sum must be even because $2m$ is even and the sum of the degrees of the vertices of even degrees is also even. Because this is the sum of the degrees of all vertices of odd degree in the graph, there must be an even number of such vertices.